



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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COMMON QUARTERLY EXAMINATION - 2024 - 25

Time Allowed : 3.00 Hours]

MATHEMATICS

[Max. Marks : 90

PART - I

20 x 1 = 20

- Answer all the questions by choosing the correct answer from the given 4 alternatives
 - Write question number, correct option and corresponding answer
 - Each question carries 1 mark
- If $A = \{(x, y) : y = \sin x, x \in \mathbb{R}\}$ and $B = \{(x, y) : y = \cos x, x \in \mathbb{R}\}$ then $A \cap B$ contains
 - no element
 - infinitely many elements
 - only one element
 - cannot be determined.
 - Let A and B be subsets of the universal set N , the set of natural numbers. Then $A' \cup [(A \cap B) \cup B']$ is
 - A
 - A'
 - B
 - N
 - The rule $f(x) = x^2$ is a bijection if the domain and the co-domain are given by
 - $\mathbb{R}, \mathbb{R}$
 - $\mathbb{R}, (0, \infty)$
 - $(0, \infty), \mathbb{R}$
 - $[0, \infty), [0, \infty)$
 - If $|x + 2| \leq 9$, then x belongs to
 - $(-\infty, -7)$
 - $[-11, 7]$
 - $(-\infty, -7) \cup [11, \infty)$
 - $(-11, 7)$
 - The solution of $5x - 1 < 24$ and $5x + 1 > -24$ is
 - $(4, 5)$
 - $(-5, -4)$
 - $(-5, 5)$
 - $(-5, 4)$
 - If $\frac{1-2x}{3+2x-x^2} = \frac{A}{3-x} + \frac{B}{x+1}$, then the value of $A + B$ is
 - $\frac{-1}{2}$
 - $\frac{-2}{3}$
 - $\frac{1}{2}$
 - $\frac{2}{3}$
 - The maximum value of $4\sin^2 x + 3\cos^2 x + \sin \frac{x}{2} + \cos \frac{x}{2}$ is
 - $4 + \sqrt{2}$
 - $3 + \sqrt{2}$
 - 9
 - 4
 - Which of the following is not true?
 - $\sin \theta = -\frac{3}{4}$
 - $\cos \theta = -1$
 - $\tan \theta = 25$
 - $\sec \theta = \frac{1}{4}$
 - If $\sin \alpha + \cos \alpha = b$, then $\sin 2\alpha$ is equal to
 - $b^2 - 1$, if $b \leq \sqrt{2}$
 - $b^2 - 1$, if $b > \sqrt{2}$
 - $b^2 - 1$, if $b \geq 1$
 - $b^2 - 1$, if $b \geq \sqrt{2}$
 - The number of ways in which the following prize be given to a class of 30 boys first and second in mathematics, first and second in physics, first in chemistry and first in English is
 - $30^4 \times 29^2$
 - $30^3 \times 29^3$
 - $30^2 \times 29^4$
 - 30×29^5
 - There are 10 points in a plane and 4 of them are collinear. The number of straight lines joining any two points is
 - 45
 - 40
 - 39
 - 38
 - If 10 lines are drawn in a plane such that no two of them are parallel and no three are concurrent, then the total number of points of intersection are
 - 45
 - 40
 - 10!
 - 2^{10}



13. The coefficient of x^8y^{12} in the expansion of $(2x + 3y)^{20}$ is
(1) 0 (2) 2^83^{12} (3) $2^83^{12} + 2^{12}3^8$ (4) $^{20}C_82^83^{12}$
14. The sequence $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}+\sqrt{2}}, \frac{1}{\sqrt{3}+2\sqrt{2}}, \dots$ form an
(1) AP (2) GP (3) HP (4) AGP.
15. The n^{th} term of the sequence $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \dots$ is
(1) $2^n - n - 1$ (2) $1 - 2^{-n}$ (3) $2^{-n} + n - 1$ (4) 2^{n-1}
16. If the point $(8, -5)$ lies on the locus $\frac{x^2}{16} - \frac{y^2}{25} = k$, then the value of k is
(1) 0 (2) 1 (3) 2 (4) 3
17. The intercepts of the perpendicular bisector of the line segment joining $(1,2)$ and $(3,4)$ with coordinate axes are
(1) 5, -5 (2) 5, 5 (3) 5, 3 (4) 5, -4
18. The length of $\perp$ from the origin to the line $\frac{x}{3} - \frac{y}{4} = 1$, is
(1) $\frac{11}{5}$ (2) $\frac{5}{12}$ (3) $\frac{12}{5}$ (4) $-\frac{5}{12}$
19. $19.(n-1)P_r + r(n-1)P_{(r-1)} =$
(1) nP_r (2) $(n+1)P_r$ (3) nC_r (4) $(n+1)C_r$
20. Number of terms in $(x+a)^6 + (x-a)^6$ is
(2) 4 (3) 12 (3) 5 (4) 7

PART - II

1. Answer any 7 questions. 2. Each question carries 2 marks.
3. Q. NO: 30 is Compulsory.

7x2=14

21. If $n(A) = 10$ and $n(A \cap B) = 3$, find $n((A \cap B)' \cap A)$.
22. Solve $23x < 100$ when (i) x is a natural number, (ii) x is an integer.
23. Find the degree measure corresponding to the following radian measures: $\frac{7\pi}{3}$
24. Find the principal value of (i) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$
25. Find the value of $\frac{8!}{5! \times 2!}$
26. If ${}^nC_{12} = {}^nC_9$ find ${}^{21}C_n$.
27. The product of three increasing numbers in GP is 5832. If we add 6 to the second number and 9 to the third number, then resulting numbers form an AP. Find the numbers in GP.



28. If $\frac{6!}{n!} = 6$, then find the value of n .

29. Find the equation of the lines passing through the point (1,1) and the perpendicular from the origin makes an angle 60° with x -axis.

30. Find the path traced out by the point $(ct, \frac{c}{t})$ here $t \neq 0$ is the parameter and c is a constant.

PART - III

1. Answer any 7 questions. 2. Each question carries 3 marks.
3. O. NO: 40 is Compulsory.

7 x 3 = 21

31. Write the values of f at $-3, 5, 2, -1, 0$ if $f(x) = \begin{cases} x^2 + x - 5 & \text{if } x \in (-\infty, 0) \\ x^2 + 3x - 2 & \text{if } x \in (3, \infty) \\ x^2 & \text{if } x \in (0, 2) \\ x^2 - 3 & \text{otherwise} \end{cases}$

32. Solve $x = \sqrt{x+20}$ for $x \in \mathbb{R}$.

33. If $(x^{1/2} + x^{-1/2})^2 = 9/2$, then find the value of $(x^{1/2} - x^{-1/2})$ for $x > 1$.

34. Prove that $\sin 105^\circ + \cos 105^\circ = \cos 45^\circ$.

35. Find the values of (i) $\sin 18^\circ$

36. Prove that $\frac{(2n)!}{n!} = 2^n(1.3.5 \dots (2n-1))$.

37. Find the sum of all 4-digit numbers that can be formed using digits 0, 2, 5, 7, 8 without repetition?

38. Find the coefficient of x^4 in the expansion of $(1+x^3)^{50} \left(x^2 + \frac{1}{x}\right)^5$

39. Find the sum up to the 17th term of the series $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \dots$

40. Find the equation of the lines make an angle 60° with positive x -axis and at a distance $5\sqrt{2}$ units measured from the point (4, 7), along the line $x - y + 3 = 0$.

PART - IV

1. Answer all the questions. 2. Each question carries 5 marks.

7 x 5 = 35

41. a) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 2x - |x|$ and $g(x) = 2x + |x|$. Find $f \circ g$.

(OR)

b) A 150m long train is moving with constant velocity of 12.5 m/s. Find (i) the equation of the motion of the train, (ii) time taken to cross a pole. (iii) The time taken to cross the bridge of length 850m is?

42. a) In the set $\mathbb{Z}$ of integers, define mRn if $m - n$ is divisible by 7. Prove that R is an equivalence relation.

(OR)



b) Prove that $C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 = \frac{2n!}{(n!)^2}$.

43. a) Resolve the following rational expressions into partial fractions : $\frac{x}{(x-1)^3}$

(OR)

b) Prove that $\sqrt[3]{x^3 + 7} - \sqrt[3]{x^3 + 4}$ is approximately equal to $\frac{1}{x^2}$ when x is large.

44. a) Prove that $\frac{\cot(180^\circ + \theta) \sin(90^\circ - \theta) \cos(-\theta)}{\sin(270^\circ + \theta) \tan(-\theta) \operatorname{cosec}(360^\circ + \theta)} = \cos^2 \theta \cot \theta$.

(OR)

b) Show that the straight lines $x^2 - 4xy + y^2 = 0$ and $x + y = 3$ form an equilateral triangle.

45. a) In a ΔABC , prove that $\sin\left(\frac{B-C}{2}\right) = \frac{b-c}{a} \cos \frac{A}{2}$.

(OR)

b) Find the image of the point $(-2, 3)$ about the line $x + 2y - 9 = 0$.

46. a) If ${}^{(n+1)}C_8 : ({}^{(n-3)}P_4) = 57 : 16$, find the value of n .

(OR)

b) Prove that $\sqrt[3]{x^3 + 7} - \sqrt[3]{x^3 + 4}$ is approximately equal to $\frac{1}{x^2}$ when x is large.

47. a) Solve $\sqrt{3} \sin \theta - \cos \theta = \sqrt{2}$

(OR)

b) Use induction to prove that $n^3 - 7n + 3$, is divisible by 3, for all natural numbers n .